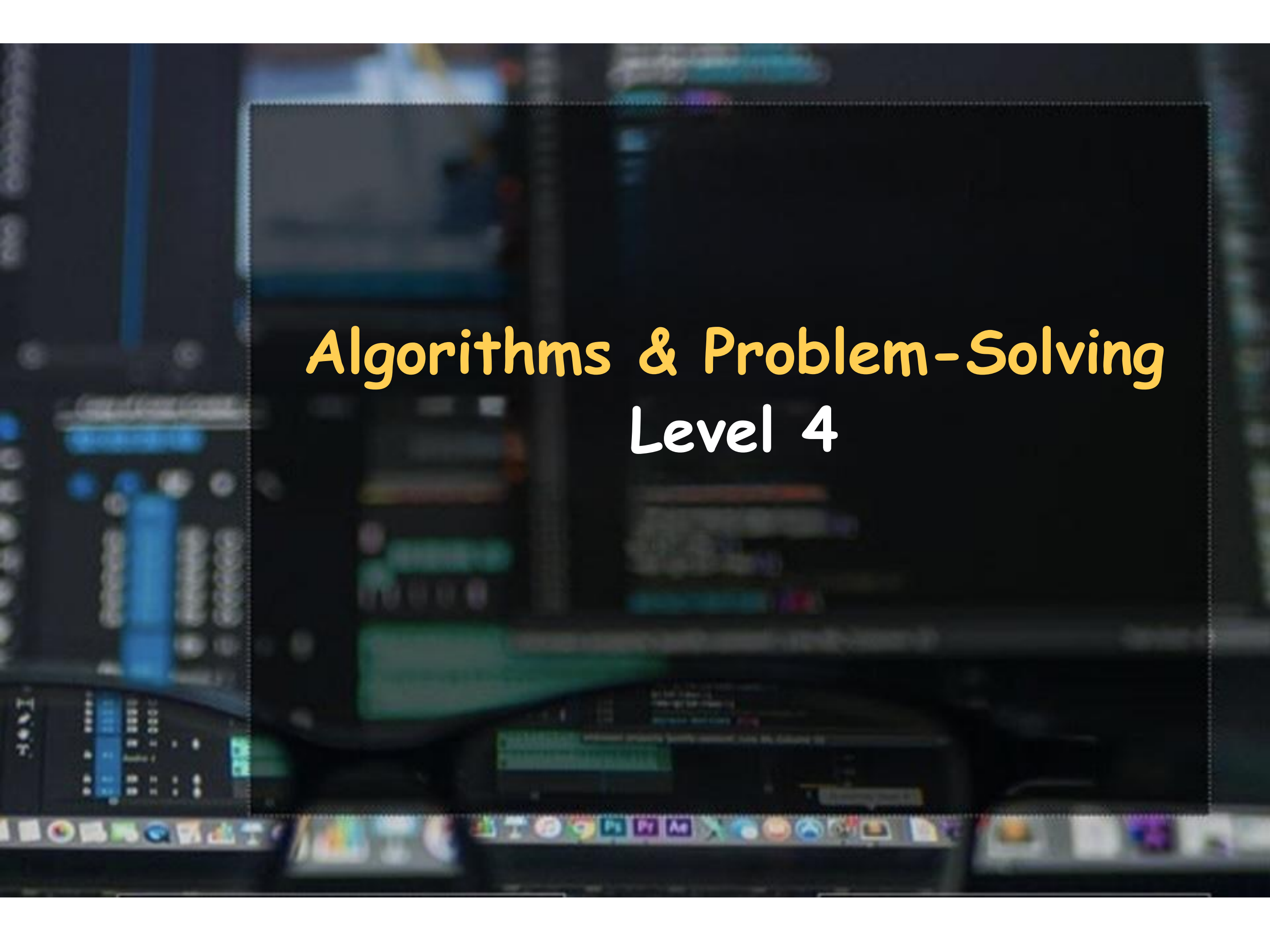
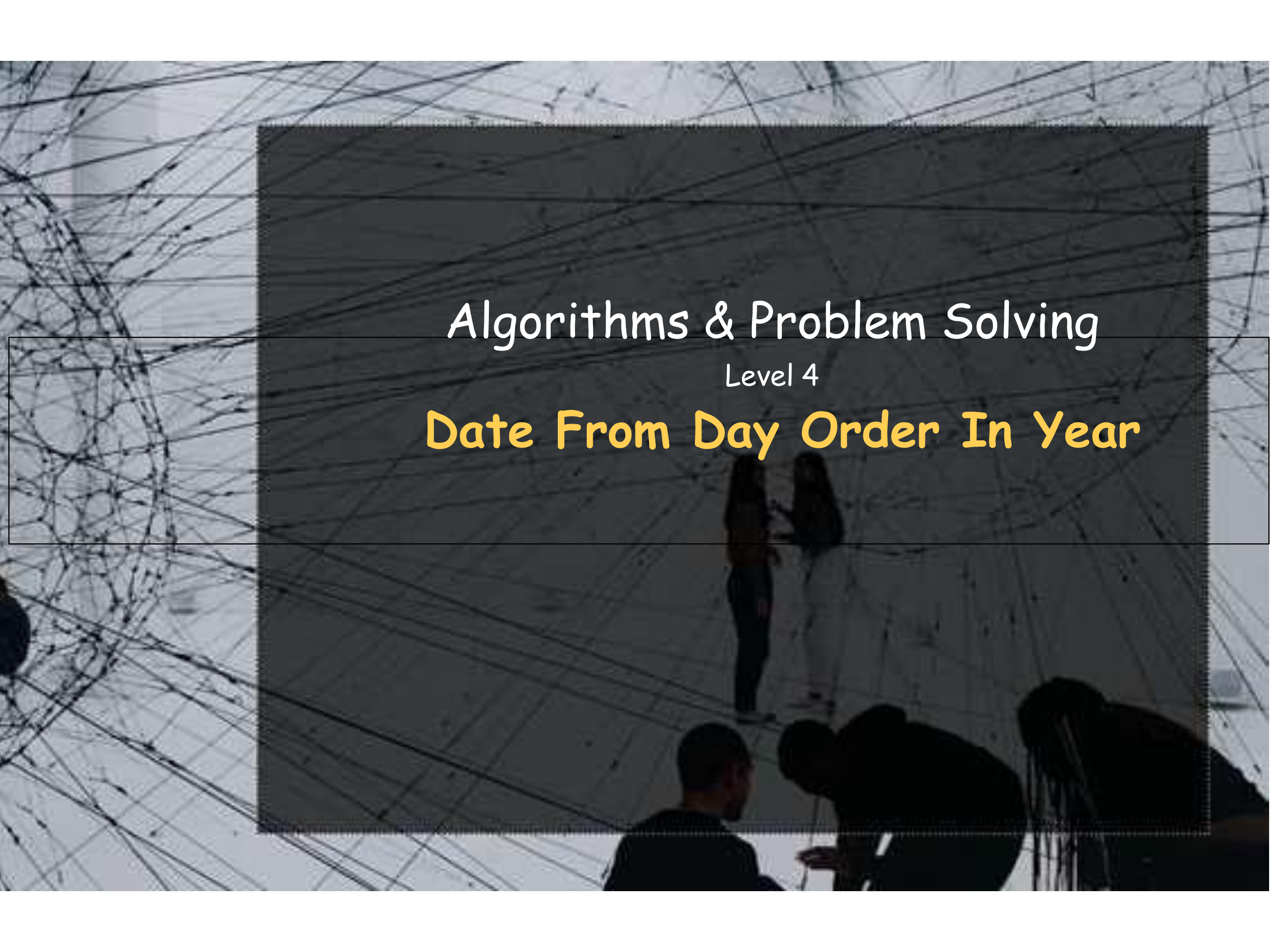




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Date From Day Order In Year



Definition

- The goal is a program that calculates the number of days since the beginning of the year to a specific date, then reconverts this number to the same date (day/month/year).



Explanation

- The program performs two main steps:
 - Step 1: Calculates the number of days since the beginning of the year using `NumberOfDaysFromTheBeginningOfTheYear`
 - Step 2: Converts this number back to an actual date using `GetDateFromDayOrderInYear`
- The program uses leap years to accurately determine the number of days in February.

Procedures

- 1 Read values from the user
 -  ReadYear() Enter the year
 -  ReadMonth() Enter the month
 -  ReadDay() Enter the day
- 2 Checking for a leap year
 -  Using the function
bool IsLeapYear(int year)
- 3 Calculate the number of days since the beginning of the year
 -  Using the function
int NumberOfDaysFromTheBeginningOfTheYear(int day, int month, int year)
- 4 Converting a Number to a Date
 -  Using the sDate structure and the function
sDate GetDateFromDayOrderInYear(int dateOrderInYear, int year)
 -  The function subtracts the days of each month from the number until the correct month is reached, then determines the day.



Example:

Please enter a Day? 20

Please enter a Month? 5

Please enter a Year? 1981

Number of Days from the beginning of the year is 140

Date for [140] is: 20/5/1981



Note:

- The program is accurate whether the year is a leap year or not.
- The sDate structure is used to store the resulting date (day, month, year).

Summary

- ✅ The program links the day of the year to the actual date and is suitable for accurately calculating and converting any date within the year.

Thank you 😊